Breathing In, Breathing Out — Paper D

HARD · 45 MINUTES · 30 MARKS

NAME DATE SCORE / 30

- Answer every question in the space provided.
- A calculator is allowed, but every number here is designed to work without one.
- Where a question asks about air moving in or out, your answer must follow the chain in order: muscles change the volume, the volume change causes a pressure change, air then flows down the pressure gradient. A muscle never sucks or pushes air by itself.
- The mark for each question is shown in square brackets on the right.
- Several questions here describe experiments and data you may not have seen before. You are not expected to recognise them — apply the ideas you already have.
- Never write that respiration takes place in the lungs. Respiration is a reaction inside cells; the lungs carry out ventilation and gas exchange.

SECTION 1 — A LUNG THAT STOPS WORKING

1. In a long-term smoker the walls between neighbouring alveoli break down, so many small alveoli merge into fewer, much larger sacs. This is emphysema. One patient's total gas exchange surface falls from 50 m^2 to 20 m^2.

(a) Calculate the percentage fall in his gas exchange surface area. [2]

(b) Explain the effect this has on gas exchange, and what the patient would notice when he climbs a flight of stairs. [3]

2. Tobacco smoke also contains carbon monoxide, which binds to haemoglobin in red blood cells far more readily than oxygen does, and does not let go.

(a) Explain why this reduces the amount of oxygen the blood can deliver, even from healthy alveoli. [2]

(b) At rest this patient's blood delivers 12% less oxygen than it should. If it should deliver 250 cm^3 of oxygen each minute, calculate how much it actually delivers. [2]

3. Tar in the smoke paralyses and then destroys the cilia lining the airways. Predict two things that happen as a result, and explain each one. [3]

4. Describe and explain the complete journey of an oxygen molecule from the air just outside a person's nose to a mitochondrion inside a muscle cell in her leg. Name the structures it passes through and explain what makes it move at each stage. [6]

Six marks means at least six separate ideas. Plan the order of the structures in the margin before you begin writing.

5. High in the mountains the air is still 21% oxygen, but the air pressure is much lower, so each breath contains fewer oxygen molecules. A climber who arrives at 4000 m breathes faster and more deeply than at sea level, even when sitting still.

(a) Explain why breathing faster and more deeply helps the climber. [2]

(b) After two weeks at altitude the climber makes more red blood cells than before. Explain how this helps. [2]

6. Complete the comparison of the two types of respiration. Each empty box is worth one mark. [3]

	Aerobic	**Anaerobic (in muscle)**
Oxygen needed?	yes	
Products	carbon dioxide + water	
Energy released		much less

7. Another student has answered two questions on this topic. Both answers are wrong. For each one, state the mistake and write out a correct version. [5]

(i) “Respiration takes place in the lungs, where oxygen is turned into carbon dioxide.”

(ii) “When you breathe out, the diaphragm pushes down on the lungs and forces the air out of them.”

Each one has more than one thing wrong with it. Deal with them separately.

End of paper. Check every answer has a unit or a form the question actually asked for.